Avisalaaaaa Art Museum: Network Design with Guest Network

1. Departments and VLAN Assignment

VLAN ID	VLAN Name	Departments / Purpose	Switch
10	Staff	HR, Finance, Security, Curation	S1
20	IT_Admin	IT, Admin, File/DB Server	S2
30	Guest	Guest visitors (wired + wireless), FTP Server	S3

Note: VLAN 1 (default) remains on all switches for management traffic only; no end devices are assigned to it.

2. IP Addressing Table

2.1 VLAN Subnets and Gateways

VLAN	Subnet	Router Sub-Interface	Gateway IP
VLAN 10 – Staff	192.168.10.0/24	Gi0/0.10 (dot1Q 10)	192.168.10.1
VLAN 20 – IT/Admin	192.168.20.0/24	Gi0/0.20 (dot1Q 20)	192.168.20.1
VLAN 30 – Guest	192.168.30.0/24	Gi0/0.30 (dot1Q 30)	192.168.30.1

2.2 Device IP Assignments

VLAN	Device	IP Address	Type
10	HR PC	192.168.10.11	DHCP
10	Finance PC	192.168.10.12	DHCP
10	Security PC	192.168.10.13	DHCP
10	Curation PC	192.168.10.14	DHCP
20	IT PC	192.168.20.11	Static
20	Admin PC	192.168.20.12	Static
20	File/DB Server	192.168.20.5	Static
30	Guest PC 1	192.168.30.10	DHCP
30	Guest PC 2	192.168.30.11	DHCP
30	Guest PC 3	192.168.30.12	DHCP
30	FTP Server	192.168.30.5	Static
30	Access Point (mgmt)	DHCP-scope	DHCP
30	Wireless client (Laptop0)	192.168.30.13 (leased)	DHCP

2.3 WAN / Router-to-Router Link

Link	Subnet	Device	IP Address
WAN uplink	203.0.113.0/30	CoreRouter Gi0/1	203.0.113.1
WAN uplink	203.0.113.0/30	InternetRouter Gi0/1	203.0.113.2

3. Devices and Port Assignments

Device	Model	Purpose	Ports Used	Cable Type
Internet Router	Cisco 2911	WAN edge, default route	Gi0/1 → CoreRouter	Copper Straight-Through
Core Router	Cisco 2911	Router-on-a-stick, DHCP, ACL, HSRP	Gi0/0 → Switch0 (trunk); Gi0/1 → InternetRouter	Copper Straight-Through
Switch0 (distribution)	Cisco 2960-24TT	Passthrough trunk to CoreRouter	Gi0/1 → CoreRouter; Gi0/2 → S1	Copper Straight-Through
S1	Cisco 2960-24TT	VLAN 10 access, trunk uplink, STP redundancy	Fa0/1–4 → Staff PCs; Gi0/1 → Switch0; Gi0/2 → S2; Fa0/23 → S3 (redundant)	Straight-Through (PCs), Cross-Over (S1↔S3)
S2	Cisco 2960-24TT	VLAN 20 access, trunk uplinks	Fa0/1–3 → IT/Admin PCs + File/DB Server; Gi0/1 → S1; Gi0/2 → S3	Copper Straight-Through
S3	Cisco 2960-24TT	VLAN 30 access, trunk uplink, STP redundancy	Fa0/1–4 → Guest PCs; Fa0/5 → Access Point; Gi0/1 → S2; Fa0/23 → S1 (redundant)	Straight-Through (PCs/AP), Cross-Over (S1↔S3)

Access Point	AP-PT	Wireless guest access, SSID + WPA2	Port0 → S3 Fa0/5	Copper Straight-Through
Wireless client	Laptop-PT	Guest wireless test device	Wireless (802.11)	N/A — wireless
FTP Server	Server-PT	Guest-facing FTP service	Fa0 → S3	Copper Straight-Through
File/DB Server	Server-PT	IT/Admin data service	Fa0 → S2	Copper Straight-Through

4. Network Design Summary

Topology: InternetRouter — CoreRouter — Switch0 — S1 — S2 — S3, with a redundant Fa0/23 cross-over link directly between S1 and S3 for Spanning Tree high availability. Each access switch (S1, S2, S3) hosts one VLAN's end devices; the Core Router performs inter-VLAN routing via router-on-a-stick (802.1Q sub-interfaces). A Wireless Access Point extends VLAN 30 to wireless clients. The Internet Router provides the WAN edge, reached via a static default route.

5. Full Configuration Commands

5.1 VLAN Creation (per switch)

S1:

```
vlan 10
 name Staff
exit
```

S2:

```
vlan 20
 name IT_Admin
exit
```

S3:

```
vlan 30
 name Guest
exit
```

Switch0 (distribution — must know all VLANs even with no local devices):

```
vlan 10
 name Staff
vlan 20
 name IT_Admin
vlan 30
 name Guest
exit
```

5.2 Access Ports (per switch)

S1 – Staff PCs:

```
interface range fastEthernet 0/1-4
 switchport mode access
 switchport access vlan 10
exit
```

S2 – IT/Admin PCs + File/DB Server:

```
interface range fastEthernet 0/1-3
 switchport mode access
 switchport access vlan 20
exit
```

S3 – Guest PCs + FTP Server:

```
interface range fastEthernet 0/1-4
 switchport mode access
 switchport access vlan 30
```

```
exit
```

S3 – Access Point:

```
interface fastEthernet 0/5
 switchport mode access
 switchport access vlan 30
exit
```

5.3 Trunk Ports (manual — DTP dynamic auto does not form trunks)

Switch0:

```
interface gig0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
interface gig0/2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
end
```

S1:

```
interface gig0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
interface gig0/2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
interface fastEthernet 0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
end
```

S2:

```
interface gig0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
interface gig0/2
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
end
```

S3:

```
interface gig0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
interface fastEthernet 0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
end
```

5.4 Core Router — Sub-interfaces and Inter-VLAN Routing

```
hostname CoreRouter
interface gigabitEthernet 0/0
 no ip address
 no shutdown
exit
interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
```

```
exit
interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
```

5.5 Internet Router Uplink and Default Route

CoreRouter:

```
interface gigabitEthernet 0/1
 ip address 203.0.113.1 255.255.255.252
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

InternetRouter:

```
hostname InternetRouter
interface gigabitEthernet 0/1
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 203.0.113.1
ip route 192.168.20.0 255.255.255.0 203.0.113.1
ip route 192.168.30.0 255.255.255.0 203.0.113.1
```

5.6 DHCP Pools

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool VLAN10_STAFF
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
exit

ip dhcp excluded-address 192.168.30.1 192.168.30.9
ip dhcp excluded-address 192.168.30.5
ip dhcp pool VLAN30_GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
exit
```

5.7 Static IP Devices

Configured directly on each device (Desktop/Config → IP Configuration → Static):

Device	IP Address	Subnet Mask	Default Gateway
IT PC	192.168.20.11	255.255.255.0	192.168.20.1
Admin PC	192.168.20.12	255.255.255.0	192.168.20.1
File/DB Server	192.168.20.5	255.255.255.0	192.168.20.1
FTP Server	192.168.30.5	255.255.255.0	192.168.30.1

5.8 Guest ACL — FTP-Only Access (final version)

Strict version: Guest VLAN may only reach its own gateway and the FTP service on the FTP server. All other Guest-initiated traffic is blocked, including replies to pings initiated by other VLANs.

```
no access-list 100
access-list 100 permit icmp 192.168.30.0 0.0.0.255 host 192.168.30.1
access-list 100 permit tcp 192.168.30.0 0.0.0.255 host 192.168.30.5 eq 21
access-list 100 permit tcp 192.168.30.0 0.0.0.255 host 192.168.30.5 eq 20
access-list 100 deny ip 192.168.30.0 0.0.0.255 any
access-list 100 permit ip any any
```

```
interface g0/0.30
 ip access-group 100 in
end
```

5.9 Spanning Tree Priorities

```
S1(config)# spanning-tree vlan 10 priority 4096
S2(config)# spanning-tree vlan 20 priority 8192
S3(config)# spanning-tree vlan 30 priority 12288
```

5.10 HSRP (First Hop Redundancy)

```
interface g0/0.10
 standby 10 ip 192.168.10.1
 standby 10 priority 110
 standby 10 preempt
interface g0/0.20
 standby 20 ip 192.168.20.1
 standby 20 priority 110
 standby 20 preempt
interface g0/0.30
 standby 30 ip 192.168.30.1
 standby 30 priority 110
 standby 30 preempt
```

Limitation: only one physical router currently serves as gateway, so all groups report State = Active with Standby = unknown. True failover requires a second router in the same HSRP groups.

5.11 Wireless Access Point

```
SSID: Avisala_GuestWiFi
Authentication: WPA2-PSK
Pass Phrase: AvisalaMuseum2026!
```

5.12 Port Security (optional — Staff and IT/Admin ports)

```
S1(config)# interface range fa0/1-4
S1(config-if-range)# switchport port-security
S1(config-if-range)# switchport port-security maximum 1
S1(config-if-range)# switchport port-security violation shutdown
S1(config-if-range)# switchport port-security mac-address sticky
```

6. Testing and Verification Steps

6.1 Gateway Reachability

From every PC/server, ping its own default gateway (192.168.10.1 / .20.1 / .30.1) — all must succeed.

6.2 Trunk and VLAN Verification

```
show interfaces trunk
show vlan brief
show ip interface brief
```

6.3 DHCP Verification

```
show ip dhcp pool
show ip dhcp binding
```

6.4 Guest ACL Verification

From any Guest PC or the wireless client:

```
ping 192.168.30.1      ! gateway — success
ping 192.168.30.5      ! FTP server — success
ping 192.168.10.11     ! HR — fail (blocked)
ping 192.168.20.5      ! File/DB Server — fail (blocked)
ftp 192.168.30.5       ! login cisco/cisco — success
```

6.5 Cross-VLAN Connectivity (Staff ↔ IT/Admin — unrestricted)

From HR/Finance/Security/Curation to IT PC/Admin PC/File-DB Server, and vice versa — all should succeed since no ACL restricts VLAN 10 ↔ VLAN 20 traffic.

6.6 Internet Router Connectivity

```
CoreRouter#ping 203.0.113.2
InternetRouter#ping 203.0.113.1
CoreRouter#show ip route
```

6.7 STP Failover Simulation

```
S3#show spanning-tree          ! confirm Fa0/23 = Altn/BLK, Gi0/1 = Root/FWD
S3(config)# interface gig0/1
S3(config-if)# shutdown
! ping from a Guest device to 192.168.30.1 during the outage — expect 0% loss
S3#show spanning-tree          ! confirm Fa0/23 transitioned to FWD
S3(config)# interface gig0/1
S3(config-if)# no shutdown     ! restore
```

6.8 Wireless Client Verification

Connect Laptop0 to SSID Avisala_GuestWiFi (WPA2-PSK), obtain DHCP address, then repeat the Guest ACL tests in 6.4 — results must match the wired Guest PCs exactly.